

EVENT REPORT

MECHNOTRON 2K25

Date: 21/03/2025

Organized by: Mechanical Engineering Association (MEA), CIT

Venue: CIT Campus, Coimbatore

Introduction

The Department of Mechanical Engineering at CIT conducted its annual national-level technical symposium MECHNOTRON 2K25 on 21st March 2025 under the banner of the Mechanical Engineering Association (MEA). The event was envisioned to bring together young minds from various engineering institutions across the country to compete, collaborate, and contribute towards innovation in mechanical and allied engineering domains. The symposium comprised a total of 21 technical and non-technical events, meticulously planned and executed to test the creativity, critical thinking, and technical expertise of participants.

With a turnout of over 1300 students from nearly 40 colleges, the event was a vibrant celebration of engineering spirit, teamwork, and innovation. Below is a detailed account of each event conducted during the symposium.

Event Summary

1. Paper Presentation

TIME: 9:00 AM - 1:00 PM

The Paper Presentation event served as a scholarly platform where students presented their research findings, innovative designs, and futuristic concepts related to mechanical engineering and interdisciplinary areas. The topics ranged from renewable energy systems, robotics, autonomous navigation, manufacturing innovations, to sustainable materials. The session encouraged intellectual exchange and was judged by a panel of experienced faculty members based on novelty, technical content, presentation clarity, and question handling. The event aimed to nurture research thinking and academic confidence among students. Approximately 160 teams actively participated in the event.



2. Hackathon

TIME:9:00 AM - 1:00 PM

The Hackathon was an intense brainstorming challenge where participants tackled real-time industrial problems or engineering issues. Students worked in teams to develop software tools or control systems with practical applications. Themes included automation in manufacturing, logistics optimization, and intelligent control systems. The event was judged on innovation, problem-solving approach, functionality of the prototype, and presentation. It provided a real-world experience in time management, team collaboration, and solution-driven engineering. Approximately 16 teams actively participated in the event.



3. Projectopia

TIME:9:00 AM - 1:00 PM

The Projectopia event allowed the students to exhibit their completed or ongoing technical projects, models, or working prototypes. The projects displayed ranged from robotic arms, hydraulic systems, home automation devices to sustainable energy solutions. Participants explained their concepts to visitors and judges, who evaluated the projects based on creativity, real-world application, cost-effectiveness, and technical execution. This event gave hands-on skills and practical engineering knowledge among the students. Approximately 20 teams actively participated in the event.



4. Designathon

TIME:9:00 AM - 1:00 PM

Designathon was a Computer-Aided Design (CAD) modeling competition where participants were given real-world design problems and asked to model mechanical parts or assemblies within a specified time. Using tools like SolidWorks and AutoCAD, students demonstrated their design skills, dimensional accuracy, and understanding of mechanical functionality. The event tested their creativity, speed, and command over software tools and was judged on correctness,

innovation, and presentation of the design. Approximately 20 teams actively participated in the event.



5. Code Craze

TIME:9:00 AM - 1:00 PM

This was a multi-level programming competition focused on solving algorithmic challenges and debugging tasks. The event tested the logical thinking, algorithm development, and coding efficiency of participants. With each level increasing in complexity, students were evaluated on time taken, accuracy of solutions, and the ability to work under pressure. The competition attracted participants from computer science, mechanical, and electronics backgrounds, proving the interdisciplinary appeal of the event. Approximately 80 students actively participated in the event.



6. Egg Drop Challenge

TIME: 2:00 PM - 4:00 PM

In this classic engineering challenge, participants were tasked with designing a protective structure for a raw egg using limited materials. The aim was to prevent the egg from breaking when dropped from a height. The event encouraged students to think about impact absorption, material behavior, and structural design. Some of the most successful designs used principles of crumple zones and shock dispersion. It was not only fun but also served as a real-time experiment in mechanical shock resistance. Approximately 10 teams actively participated in the event.

7. Aqua Propulsion

TIME: 9:00 AM - 1:00 PM

Aqua Propulsion was a hands-on event where participants constructed water-powered rockets using plastic bottles, fins, and nozzles. The challenge was to design a rocket that could fly the farthest and most accurately using only water and air pressure as propulsion. Students experimented with different fin shapes, bottle configurations, and launch angles. This event

merged theoretical knowledge with exciting experimentation and highlighted the physics of propulsion and fluid dynamics. Approximately 13 teams actively participated in the event.



8. Debate Fusion

TIME: 2:00 PM - 4:00 PM

Debate Fusion was an intellectually stimulating event where students debated on contemporary and controversial topics in the field of engineering, technology, and society. Topics included automation vs employment, nuclear energy, ethical AI, and more. Participants were evaluated on their argument strength, rebuttal skills, clarity of thought, and ability to present ideas logically. The event improved public speaking, critical analysis, and persuasive communication skills among participants. Approximately 10 teams actively participated in the event.

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9. Pitch Masters

TIME: 9:30 AM– 11:30AM

This entrepreneurial event provided a stage for students to present startup ideas or innovative business models to a panel of evaluators acting as investors. Students pitched their ideas involving engineering innovations, sustainability products, and problem-solving platforms. Each team had to cover the product's need, market scope, business strategy, and scalability. This event helped students understand business modeling, marketing strategy, and real-world startup challenges. Approximately 5 teams actively participated in the event.

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10. Deconstructx

TIME: 02:00PM –03:30PM

Deconstructx event tested the analytical and practical abilities of students to disassemble mechanical systems, understand their working, and recreate them or sketch their internal components. Participants were provided with components from common appliances or machine parts. They were judged based on their understanding of function, part integration, and ability to document the re-engineered version accurately. The event emphasized observation, logical deduction, and mechanical comprehension. Approximately 5 teams actively participated in the event.

11. Equilibrium Quest

TIME: 02:00PM –03:30PM

This event was a hands-on problem-solving competition where participants had to build structures or systems using given materials that could withstand loads or balance under specific constraints. The objective was to demonstrate principles of equilibrium, structural integrity, and distribution of forces. Students applied theoretical concepts like center of gravity, moment balance, and stability in a creative and engaging environment. Approximately 15 teams actively participated in the event.

12. Tech Linkx

TIME: 02:00PM –03:30PM

Tech Linkx was a fast-paced word-association game where teams had to connect mechanical engineering terminologies, components, or concepts based on clues provided. It tested students' vocabulary, technical recall, and ability to draw conceptual links between seemingly unrelated topics. The game-like format made it highly engaging while reinforcing domain knowledge in a fun way. Approximately 15 teams actively participated in the event.

13. Lathe Masters

TIME: 02:00PM – 04:30PM

Lathe Masters was a skill-based workshop event conducted in the Machine Shop. Participants performed precision operations such as turning, facing, taper cutting, and grooving on a lathe machine. The event focused on hands-on machining skills, tool control, safety protocol, and final product accuracy. Judges assessed the finish quality, dimensional accuracy, and time

management. It provided students an opportunity to demonstrate practical shop floor competence. Approximately 30 students actively participated in the event.



14. Weldx

Time: 02:00PM – 04:30PM

The Arc Welding Event aimed to challenge participants on welding precision, technique, and safety skills using industry-grade equipment. Around 30 students participated, completing a specified welding task under expert supervision. The activity evaluated weld strength, finish, and adherence to safety protocols in a time-bound setup. Participants demonstrated real-world fabrication techniques. The event successfully assessed the integrity of welded joints and highlighted individuals with technical precision, strong craftsmanship, and time-efficient execution.



15. Chess

Time: 02:00PM – 04:30PM

This chess event focused on fostering strategic thinking, patience, and decision-making skills. Around 20 participants engaged in timed matches that tested logical reasoning, mental discipline, and tactical gameplay. The competition promoted sportsmanship, focus, and resilience under pressure. It provided a platform for intellectual challenge and recognized players for their mental agility and strategic prowess.



16. Carrom

Time: 9:00AM–04:30PM

The Carrom event encouraged concentration, coordination, and precision through competitive gameplay. With around 25 teams participating, matches were held in a knockout format to maintain excitement. Players demonstrated focus, hand-eye coordination, and strategic play under timed conditions. The event emphasized teamwork, sportsmanship, and accurate shots. It successfully promoted healthy competition, improved decision-making under pressure, and identified top players with excellent aim and tactical skill.



17. Word Blitz

Time: 03:00PM – 04:30PM

Word Blitz combined vocabulary skills and basic engineering knowledge in a timed word puzzle and crossword challenge. Around 10 participants competed, displaying creativity, quick thinking, and linguistic precision. The event encouraged problem-solving under pressure while testing both language and technical knowledge. It successfully sharpened participants' technical vocabulary, mental agility, and cross-disciplinary thinking, recognizing those who excelled in both engineering and linguistic domains.

18. Minute Marvels

Time: 03:00PM – 04:30PM

Minute Marvels was a fast-paced JAM (Just A Minute) challenge testing fluency, spontaneity, and presence of mind. About 10 participants spoke on given topics for one minute without hesitation, repetition, or deviation. The event promoted confident public speaking, creative expression, and quick thinking under pressure. It provided an interactive platform that enhanced verbal articulation and rewarded those with clarity, confidence, and sharp on-the-spot delivery.

Conclusion

Mechnotron 2K25 stood as a testimony to the technical competence, creativity, and leadership of the Mechanical Engineering Association (MEA) and the student community of CIT. Each event was thoughtfully designed to nurture different facets of engineering excellence — be it coding, design, entrepreneurship, or public speaking. The symposium successfully created a space for knowledge-sharing, competition, and innovation. The overwhelming participation, smooth execution, and encouraging feedback marked the event as a grand success.

The Mechanical Engineering Association thanks all participants, faculty coordinators, student volunteers, sponsors, and the management of CIT for their unwavering support in making Mechnotron 2K25 a milestone event.

